LiDAR Sensor



Category: **Drones & Aviation** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Hovering function using LiDAR sensor

What it is used for

A LiDAR sensor measures distance using laser light and can provide height or obstacle data for drone positioning and hovering systems. Mount and configure it according to the flight controller's specifications.

How to use

1. Inspect the LiDAR sensor, lens or optical window, cables, and mounting points.
2. Confirm compatibility with the flight controller, interface, voltage, and software version.
3. Switch off the drone and disconnect the battery before wiring or mounting the sensor.
4. Mount the sensor securely in the specified orientation with a clear sensing path.
5. Connect power and data lines according to the wiring diagram.
6. Power on the system and verify sensor readings and calibration status using approved software.
7. Test hovering functions in a controlled, obstacle-free area under trained supervision.

⚠ Safety

- Keep the drone powered off while making electrical connections.
- Never look directly into the LiDAR laser aperture or aim it at people's eyes.
- Do not rely on LiDAR alone for safe flight or obstacle avoidance.
- Follow drone operating rules and maintain a safe test area.

Care & storage

- Keep the optical window clean, protect cables from strain, and check mounting security before each flight.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.